

## Volume Iv Mathematical Spaces Reference

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# Mathematical Spaces

## 0.1 Spaces as Structured Arenas

**Definition 0.1.1** (Space). A **space** is a nonempty set.

### Proofs

# Metric Spaces

## 0.2 Metric Spaces

### 0.3 Definitions

**Definition 0.3.1** (Metric Space). Let  $X$  be a nonempty set. A function

$$d : X \times X \rightarrow \mathbb{R}_{\geq 0}$$

is called a *metric* on  $X$  if for all  $x, y, z \in X$ ,

1.  $d(x, y) \geq 0$ , and  $d(x, y) = 0$  if and only if  $x = y$  (*positive definiteness*);
2.  $d(x, y) = d(y, x)$  (*symmetry*);
3.  $d(x, y) \leq d(x, z) + d(z, y)$  (*triangle inequality*).

A *metric space* is a pair  $(X, d)$ , where  $X$  is a nonempty set and  $d$  is a metric on  $X$ .

**Definition 0.3.2** (Pseudo-Metric). Let  $X$  be a nonempty set. A function

$$d : X \times X \rightarrow \mathbb{R}_{\geq 0}$$

is called a *pseudo-metric* on  $X$  if for all  $x, y, z \in X$ ,

1.  $d(x, y) \geq 0$ , and  $d(x, x) = 0$  (*positive semi-definiteness*);
2.  $d(x, y) = d(y, x)$  (*symmetry*);
3.  $d(x, y) \leq d(x, z) + d(z, y)$  (*triangle inequality*).

**Lemma 0.3.3** (Metrics are Pseudo-Metrics). *Every metric on a nonempty set  $X$  is a pseudo-metric on  $X$ . The converse is false: there exist pseudo-metrics that are not metrics.*

*Go to proof.*

## 0.4 Metrics

### 0.4.1 Norms Induce Metrics

**Definition 0.4.1** (Norm). Let  $V$  be a real vector space. A *norm* on  $V$  is a function

$$\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$$

such that for all  $x, y \in V$  and all  $\lambda \in \mathbb{R}$ ,

1.  $\|x\| = 0$  if and only if  $x = 0$  (*positive definiteness*);

2.  $\|\lambda x\| = |\lambda| \|x\|$  (absolute homogeneity);
3.  $\|x + y\| \leq \|x\| + \|y\|$  (triangle inequality).

**Theorem 0.4.2** (A Norm Induces a Metric). *Let  $V$  be a real vector space and let  $\|\cdot\|$  be a norm on  $V$ . Define*

$$d_{\|\cdot\|} : V \times V \rightarrow \mathbb{R}_{\geq 0}, \quad d_{\|\cdot\|}(x, y) := \|x - y\|.$$

*Then  $d_{\|\cdot\|}$  is a metric on  $V$ .*

*Go to proof.*

**Definition 0.4.3** (Discrete Metric). Let  $X$  be a nonempty set. The *discrete metric* on  $X$  is the function

$$d_{\text{disc}} : X \times X \rightarrow \mathbb{R}_{\geq 0}$$

defined by

$$d_{\text{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

**Proposition 0.4.4** (The Discrete Distance is a Metric). *For every nonempty set  $X$ , the discrete distance  $d_{\text{disc}}$  is a metric on  $X$ .*

*Go to proof.*

**Definition 0.4.5** (Euclidean Distance on  $\mathbb{R}$ ). The *Euclidean distance* on  $\mathbb{R}$  is the function

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}, \quad d_{\mathbb{R}}(x, y) = |x - y|.$$

**Proposition 0.4.6** (The Real-Line Euclidean Distance is a Metric). *The function  $d_{\mathbb{R}}(x, y) = |x - y|$  is a metric on  $\mathbb{R}$ .*

*Go to proof.*

**Definition 0.4.7** (Euclidean Space). For  $n \in \mathbb{N}$ , the  *$n$ -dimensional Euclidean space* is  $\mathbb{R}^n$  equipped with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

**Definition 0.4.8** (Euclidean Distance on  $\mathbb{R}^n$ ). For  $x = (x_1, \dots, x_n)$  and  $y = (y_1, \dots, y_n)$  in  $\mathbb{R}^n$ , the *Euclidean distance* is the function

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}, \quad d_2(x, y) = \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}.$$

**Theorem 0.4.9** (Cauchy–Schwarz Inequality). *For every  $x, y \in \mathbb{R}^n$ ,*

$$|x \cdot y| \leq |x| |y|.$$

*Equality holds if and only if  $x$  and  $y$  are linearly dependent.*

*Go to proof.*

**Corollary 0.4.10** (Minkowski's Inequality). *For every  $x, y \in \mathbb{R}^n$ ,*

$$|x + y| \leq |x| + |y|.$$

*Go to proof.*

**Proposition 0.4.11** (The Euclidean Distance on  $\mathbb{R}^n$  is a Metric). *The function  $d_2$  is a metric on  $\mathbb{R}^n$ .*

*Go to proof.*

**Corollary 0.4.12** (Euclidean Metric). *For every  $x, y \in \mathbb{R}^n$ , define*

$$d_2(x, y) := \sqrt{|x_1 - y_1|^2 + \dots + |x_n - y_n|^2}.$$

*Then  $(\mathbb{R}^n, d_2)$  is a metric space.*

*Go to proof.*

**Definition 0.4.13** (Taxicab Metric). For  $x, y \in \mathbb{R}^n$ , the *taxicab metric* is the function

$$d_1(x, y) := \sum_{i=1}^n |x_i - y_i|.$$

**Proposition 0.4.14** (Taxicab Metric). *The function  $d_1$  is a metric on  $\mathbb{R}^n$ .*

*Go to proof.*

**Definition 0.4.15** ( $p$ -Metric on  $\mathbb{R}^n$ ). Let  $1 \leq p < \infty$ . For  $x, y \in \mathbb{R}^n$ , define

$$d_p(x, y) := \left( \sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}.$$

**Theorem 0.4.16** ( $d_p$  is a Metric). *For every  $1 \leq p < \infty$ , the function  $d_p$  is a metric on  $\mathbb{R}^n$ .*

*Go to proof.*

**Definition 0.4.17** (Complex Metric). Identify  $\mathbb{C}$  with  $\mathbb{R}^2$ . The *complex metric* is

$$d_{\mathbb{C}}(z, w) := |z - w|.$$

**Proposition 0.4.18** (The Complex Plane is a Metric Space). *The pair  $(\mathbb{C}, d_{\mathbb{C}})$  is a metric space.*

*Go to proof.*

**Definition 0.4.19** (Supremum Metric on Bounded Real Sequences). Let  $\ell^\infty(\mathbb{R})$  denote the set of all bounded real sequences:

$$\ell^\infty(\mathbb{R}) := \{x = (x_n)_{n \in \mathbb{N}} : \exists M \geq 0 \forall n \in \mathbb{N} |x_n| \leq M\}.$$

For  $x, y \in \ell^\infty(\mathbb{R})$ , define

$$d_\infty(x, y) := \sup_{n \in \mathbb{N}} |x_n - y_n|.$$

**Proposition 0.4.20** (The Supremum Distance on Bounded Sequences is a Metric). *The function  $d_\infty$  is a metric on  $\ell^\infty(\mathbb{R})$ .*

*Go to proof.*

## 0.4.2 Point Functions and Pointlike Functions

**Definition 0.4.21** (Point Function). Let  $(X, d)$  be a metric space and let  $z \in X$ . The *point function at  $z$*  is the function

$$\delta_z : X \rightarrow \mathbb{R}_{\geq 0}, \quad \delta_z(x) := d(z, x).$$

The set of all point functions on  $X$  is denoted by

$$\delta(X) := \{\delta_z : z \in X\}.$$

**Theorem 0.4.22** (Point Functions Identify Points). *Let  $(X, d)$  be a metric space. The map*

$$X \rightarrow \delta(X), \quad z \mapsto \delta_z$$

*is a bijection.*

*Go to proof.*

**Theorem 0.4.23** (Point Function Inequalities). *Let  $(X, d)$  be a nonempty metric space and let  $z \in X$ . Then, for all  $a, b \in X$ ,*

$$\delta_z(b) - \delta_z(a) \leq d(a, b) \leq \delta_z(b) + \delta_z(a),$$

*and*

$$\delta_z(z) = 0.$$

*Go to proof.*

**Definition 0.4.24** (Pointlike Function). Let  $(X, d)$  be a nonempty metric space. A function

$$u : X \rightarrow \mathbb{R}_{\geq 0}$$

is *pointlike* on  $X$  if, for all  $a, b \in X$ ,

$$u(a) - u(b) \leq d(a, b) \leq u(a) + u(b).$$

**Theorem 0.4.25** (Pointlike Functions with a Zero Are Point Functions). *Let  $(X, d)$  be a metric space and let  $u : X \rightarrow \mathbb{R}_{\geq 0}$ . Then  $u \in \delta(X)$  if and only if  $u$  is pointlike on  $X$  and*

$$0 \in u(X).$$

*In that case, there is a unique  $w \in X$  such that  $u(w) = 0$ , and*

$$u = \delta_w.$$

*Go to proof.*

## 0.5 Examples of Metric Spaces

**Definition 0.5.1** (Discrete Metric Space). Let  $X$  be a nonempty set. The *discrete metric space* on  $X$  is the metric space

$$(X, d_{\text{disc}}),$$

where  $d_{\text{disc}}$  is the discrete metric on  $X$ .

**Definition 0.5.2** (Euclidean Metric Space). For  $n \in \mathbb{N}$ , the *standard Euclidean metric space* is

$$(\mathbb{R}^n, d_2),$$

where  $d_2$  is the Euclidean metric on  $\mathbb{R}^n$ .

**Definition 0.5.3** (Taxicab Metric Space). For  $n \in \mathbb{N}$ , the *taxicab metric space* is

$$(\mathbb{R}^n, d_1),$$

where  $d_1$  is the taxicab metric on  $\mathbb{R}^n$ .

**Definition 0.5.4** ( $p$ -Metric Space). Let  $1 \leq p < \infty$  and let  $n \in \mathbb{N}$ . The  *$p$ -metric space* is

$$(\mathbb{R}^n, d_p),$$

where  $d_p$  is the  $p$ -metric on  $\mathbb{R}^n$ .

**Definition 0.5.5** (Complex Metric Space). The *complex metric space* is

$$(\mathbb{C}, d_{\mathbb{C}}),$$

where  $d_{\mathbb{C}}$  is the usual Euclidean distance on the complex plane.

**Definition 0.5.6** (Bounded Sequence Metric Space). The *bounded sequence metric space* is

$$(\ell^\infty(\mathbb{R}), d_\infty),$$

where  $\ell^\infty(\mathbb{R})$  is the set of bounded real sequences and  $d_\infty$  is the supremum metric.

## 0.6 Distances, Boundedness, and Balls

### 0.6.1 Distances and Boundedness

**Definition 0.6.1** (Diameter). Let  $(X, d)$  be a metric space and let  $A \subseteq X$  be nonempty. The *diameter* of  $A$  is

$$\text{diam}(A) := \sup\{d(a, b) : a, b \in A\},$$

whenever this supremum is finite, and is  $+\infty$  otherwise.

**Definition 0.6.2** (Distance from a Point to a Set). Let  $(X, d)$  be a metric space, let  $x \in X$ , and let  $A \subseteq X$  be nonempty. The *distance from  $x$  to  $A$*  is

$$d(x, A) := \inf\{d(x, a) : a \in A\}.$$

**Definition 0.6.3** (Distance Between Sets). Let  $(X, d)$  be a metric space, and let  $A, B \subseteq X$  be nonempty. The *distance between  $A$  and  $B$*  is

$$d(A, B) := \inf\{d(a, b) : a \in A, b \in B\}.$$

**Theorem 0.6.4** (Distance to a Union). *Let  $(X, d)$  be a metric space, let  $x \in X$ , and let  $A, B \subseteq X$  be nonempty. Then*

$$d(x, A \cup B) = \min\{d(x, A), d(x, B)\}.$$

*Go to proof.*

**Theorem 0.6.5** (Distance to an Intersection Bound). *Let  $(X, d)$  be a metric space, let  $x \in X$ , and let  $A, B \subseteq X$  have nonempty intersection. Then*

$$d(x, A \cap B) \geq \max\{d(x, A), d(x, B)\}.$$

*Go to proof.*

**Definition 0.6.6** (Metric Bounded Set). Let  $(X, d)$  be a metric space and let  $A \subseteq X$ . The set  $A$  is *bounded* if there exist  $x \in X$  and  $R > 0$  such that

$$d(x, a) \leq R$$

for every  $a \in A$ .

**Theorem 0.6.7** (Equivalent Boundedness Criterion). *Let  $(X, d)$  be a metric space and let  $A \subseteq X$ . Then  $A$  is bounded if and only if there exist  $x \in X$  and  $R > 0$  such that*

$$d(x, a) < R$$

*for every  $a \in A$ .*

**Theorem 0.6.8** (Center Independence). *Let  $(X, d)$  be a metric space. If  $A \subseteq X$  and there exist  $x \in X$  and  $R > 0$  such that  $d(x, a) \leq R$  for every  $a \in A$ , then for every  $y \in X$ ,*

$$d(y, a) \leq R + d(x, y)$$

*for every  $a \in A$ . Thus boundedness does not depend on the chosen center.*

**Theorem 0.6.9** (Subsets of Bounded Sets Are Bounded). *Let  $(X, d)$  be a metric space. If  $A \subseteq B \subseteq X$  and  $B$  is bounded, then  $A$  is bounded.*

**Theorem 0.6.10** (Finite Sets Are Bounded). *Every finite subset of a metric space is bounded.*



**Theorem 0.6.11** (Finite Unions of Bounded Sets Are Bounded). *Let  $(X, d)$  be a metric space, let  $n \in \mathbb{N}$ , and let  $A_1, \dots, A_n \subseteq X$ . If  $A_1, \dots, A_n$  are bounded, then*

$$\bigcup_{k=1}^n A_k$$

*is bounded.*

**Theorem 0.6.12** (Boundedness via Diameter). *Let  $(X, d)$  be a metric space and let  $A \subseteq X$  be nonempty. Then  $A$  is bounded if and only if there exists  $C > 0$  such that*

$$d(a, b) \leq C$$

*for all  $a, b \in A$ .*

**Theorem 0.6.13** (Closure of a Bounded Set Is Bounded). *Let  $(X, d)$  be a metric space. If  $A \subseteq X$  is bounded, then  $\text{cl}(A)$  is bounded.*

**Theorem 0.6.14** (Compact Sets Are Bounded). *Every compact subset of a metric space is bounded.*

**Theorem 0.6.15** (Totally Bounded Sets Are Bounded). *Every totally bounded subset of a metric space is bounded.*

## 0.6.2 Balls and Local Neighborhoods

**Definition 0.6.16** (Open Ball). *Let  $(X, d)$  be a metric space, let  $x \in X$ , and let  $r > 0$ . The *open ball* of radius  $r$  centered at  $x$  is*

$$B_d(x; r) := \{ y \in X : d(y, x) < r \}.$$

**Definition 0.6.17** (Closed Ball). *Let  $(X, d)$  be a metric space, let  $x \in X$ , and let  $r > 0$ . The *closed ball* of radius  $r$  centered at  $x$  is*

$$\overline{B}_d(x; r) := \{ y \in X : d(y, x) \leq r \}.$$

**Lemma 0.6.18** (Key Local Ball Lemma). *Let  $(X, d)$  be a metric space. If  $y \in B_d(x; r)$ , then*

$$B_d(y; r - d(x, y)) \subseteq B_d(x; r).$$

**Theorem 0.6.19** (Ball Nesting). *Let  $(X, d)$  be a metric space, let  $x \in X$ , and let  $0 < r \leq s$ . Then*

$$B_d(x; r) \subseteq B_d(x; s).$$

*If  $r < s$ , then also*

$$\overline{B}_d(x; r) \subseteq B_d(x; s).$$

**Theorem 0.6.20** (Ball Containment by Center Distance). *Let  $(X, d)$  be a metric space, let  $x, y \in X$ , and let  $r, s > 0$ .*

*If*

$$d(x, y) + r \leq s,$$

*then*

$$B_d(x; r) \subseteq B_d(y; s).$$

*If*

$$d(x, y) + r < s,$$

*then*

$$\overline{B}_d(x; r) \subseteq B_d(y; s).$$

**Theorem 0.6.21** (Intersecting Balls Imply Center-Distance Bound). *Let  $(X, d)$  be a metric space, let  $x, y \in X$ , and let  $r, s > 0$ . If*

$$B_d(x; r) \cap B_d(y; s) \neq \emptyset,$$

*then*

$$d(x, y) < r + s.$$

**Theorem 0.6.22** (Closed Balls Contain Open Balls). *Let  $(X, d)$  be a metric space. For every  $x \in X$  and every  $r > 0$ ,*

$$B_d(x; r) \subseteq \overline{B}_d(x; r).$$

## 0.7 Open and Closed Sets

**Definition 0.7.1** (Metric Open Set). *Let  $(X, d)$  be a metric space. A subset  $E \subseteq X$  is open in  $(X, d)$  if either  $E = \emptyset$  or, for every  $x \in E$ , there exists  $\varepsilon > 0$  such that*

$$B_d(x; \varepsilon) \subseteq E.$$

**Theorem 0.7.2** (Open Balls are Open). *Let  $(X, d)$  be a metric space. For every  $x \in X$  and every  $r > 0$ , the open ball  $B_d(x; r)$  is open in  $(X, d)$ .*

*Go to proof.*

**Theorem 0.7.3** (Metric Open Set Closure Properties). *Let  $X$  be a metric space. Then:*

1. *the empty set  $\emptyset$  and the space  $X$  are open sets;*
2. *an arbitrary union of open subsets of  $X$  is open;*
3. *any finite intersection of open subsets of  $X$  is open.*

*Go to proof.*

**Definition 0.7.4** (Metric Closed Set). *Let  $(X, d)$  be a metric space. A subset  $E \subseteq X$  is closed in  $(X, d)$  if its complement  $E^c = X \setminus E$  is open in  $(X, d)$ .*

**Theorem 0.7.5** (Complement of a Closed Ball Is Open). *Let  $(X, d)$  be a metric space. For every  $x \in X$  and every  $r \geq 0$ ,*

$$X \setminus \overline{B}_d(x; r) = \{ y \in X : d(x, y) > r \}$$

*is open.*

**Theorem 0.7.6** (Closed Balls Are Closed). *Let  $(X, d)$  be a metric space. For every  $x \in X$  and every  $r \geq 0$ , the closed ball*

$$\overline{B}_d(x; r) = \{ y \in X : d(x, y) \leq r \}$$

*is closed.*

**Definition 0.7.7** (Metric Closure). *Let  $(X, d)$  be a metric space and let  $E \subseteq X$ . The closure of  $E$ , denoted  $\text{cl}(E)$ , is*

$$\text{cl}(E) = \{ x \in X : \forall \varepsilon > 0, B_d(x; \varepsilon) \cap E \neq \emptyset \}.$$

**Theorem 0.7.8** (Closure of an Open Ball Is Contained in the Closed Ball). *Let  $(X, d)$  be a metric space. For every  $x \in X$  and every  $r > 0$ ,*

$$\text{cl}(B_d(x; r)) \subseteq \overline{B}_d(x; r).$$

**Definition 0.7.9** (Dense Subset). *Let  $(X, d)$  be a metric space and let  $A \subseteq X$ . The set  $A$  is dense in  $X$  if*

$$\text{cl}(A) = X.$$

**Definition 0.7.10** (Metric Interior Point). Let  $(X, d)$  be a metric space, let  $E \subseteq X$ , and let  $x \in X$ . The point  $x$  is an *interior point* of  $E$  if there exists  $\delta > 0$  such that

$$B_d(x; \delta) \subseteq E.$$

**Definition 0.7.11** (Metric Interior). Let  $(X, d)$  be a metric space and let  $E \subseteq X$ . The *interior* of  $E$  is the set of all interior points of  $E$ . It is denoted by

$$E^\circ.$$

**Theorem 0.7.12** (Interior is the Largest Open Subset). *Let  $E$  be a subset of a metric space  $X$ . Then  $E^\circ$  is the largest open subset of  $X$  contained in  $E$ . In other words:*

1.  $E^\circ \subseteq E$ ;
2.  $E^\circ$  is open;
3.  $E^\circ \supseteq O$ , for every open set  $O \subseteq E$ .

*Go to proof.*

**Definition 0.7.13** (Metric Boundary). Let  $(X, d)$  be a metric space and let  $E \subseteq X$ . The *boundary* of  $E$  is

$$\partial E = \text{cl}(E) \cap \text{cl}(X \setminus E).$$

## 0.8 Limit Points and Isolated Points

**Definition 0.8.1** (Metric Limit Point and Isolated Point). Let  $E$  be a subset of a metric space  $(X, d)$ , and let  $x \in X$ .

1. The point  $x$  is a *limit point* of  $E$  if, for every  $\varepsilon > 0$ , the ball  $B_d(x; \varepsilon)$  contains a point of  $E \setminus \{x\}$ .
2. The point  $x$  is an *isolated point* of  $E$  if  $x \in E$  and  $x$  is not a limit point of  $E$ .

**Theorem 0.8.2** (Closed Sets Contain Their Limit Points). *Let  $X$  be a metric space and let  $E \subseteq X$ . Then  $E$  is closed if and only if  $E' \subseteq E$ .*

*Go to proof.*

**Theorem 0.8.3** (Sequential Characterization of Limit Points). *Let  $(X, d)$  be a metric space, let  $E \subseteq X$ , and let  $x \in X$ . Then  $x \in E'$  if and only if there exists a sequence of distinct terms in  $E$  that converges to  $x$ .*

*Go to proof.*

**Corollary 0.8.4** (Neighborhood Characterization of Limit Points). *Let  $(X, d)$  be a metric space, let  $E \subseteq X$ , and let  $x \in X$ . Then  $x \in E'$  if and only if every neighborhood of  $x$  contains infinitely many points of  $E$ .*

*Go to proof.*

**Theorem 0.8.5** (Bolzano–Weierstrass for Bounded Infinite Subsets). *Every bounded infinite subset of  $\mathbb{R}^n$  has a limit point in  $\mathbb{R}^n$ .*

*Go to proof.*

**Proposition 0.8.6** (The Derived Set is Closed). *The set of limit points of any set is closed.*

*Go to proof.*

## 0.9 Sequential Convergence

**Definition 0.9.1** (Metric Sequential Convergence). Let  $(X, d)$  be a metric space. A sequence  $\{x_n\}$  in  $X$  is *convergent in  $X$*  if there exists  $x_0 \in X$  such that for every  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$d(x_n, x_0) < \varepsilon \quad \text{for all } n \geq N.$$

In this case,  $\{x_n\}$  converges to  $x_0$ , written

$$x_n \rightarrow x_0, \quad x_0 = \lim_{n \rightarrow \infty} x_n.$$

**Definition 0.9.2** (Metric Neighborhood). Let  $(X, d)$  be a metric space and let  $x \in X$ . A subset  $U \subseteq X$  is a *neighborhood of  $x$*  if there exists  $\delta > 0$  such that

$$B_d(x; \delta) \subseteq U.$$

**Definition 0.9.3** (Epsilon Neighborhood). Let  $(X, d)$  be a metric space, let  $x \in X$ , and let  $\varepsilon > 0$ . The  $\varepsilon$ -neighborhood of  $x$  is the open ball

$$B_d(x; \varepsilon) = \{y \in X : d(y, x) < \varepsilon\}.$$

**Theorem 0.9.4** (Metric Neighborhood Criterion for Sequential Convergence). *Let  $(X, d)$  be a metric space, let  $\{x_n\}$  be a sequence in  $X$ , and let  $x \in X$ . Then*

$$x_n \rightarrow x$$

*if and only if every neighborhood of  $x$  contains all but finitely many terms of  $\{x_n\}$ .*

*Go to proof.*

**Definition 0.9.5** (Metric Cauchy Sequence). Let  $(X, d)$  be a metric space. A sequence  $\{x_n\}$  in  $X$  is *Cauchy* if for every  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$d(x_n, x_m) < \varepsilon \quad \text{for all } n, m \geq N.$$

**Theorem 0.9.6** (Metric Convergent Sequences Have Unique Limits). *In metric spaces, convergent sequences have unique limits.*

*Go to proof.*

**Theorem 0.9.7** (Metric Cauchy Sequence with Convergent Subsequence). *Let  $\{x_n\}$  be a Cauchy sequence in a metric space  $(X, d)$ , let  $x \in X$ , and let  $\{x_{n_k}\}$  be a subsequence of  $\{x_n\}$  such that*

$$\lim_{k \rightarrow \infty} x_{n_k} = x.$$

*Then  $x_n \rightarrow x$ .*

*Go to proof.*

**Theorem 0.9.8** (Coordinatewise Sequences in  $\mathbb{R}^m$ ). *Let  $\{x_n\}$  be a sequence in  $\mathbb{R}^m$ , and let  $x_0 \in \mathbb{R}^m$ . Write*

$$x_n = (x_n^{(1)}, \dots, x_n^{(m)}) \quad \text{for all } n \in \mathbb{N} \cup \{0\}.$$

*Then:*

1.  $x_n \rightarrow x_0$  if and only if  $x_n^{(j)} \rightarrow x_0^{(j)}$  for every  $j = 1, \dots, m$ .
2.  $\{x_n\}$  is Cauchy if and only if  $\{x_n^{(j)}\}$  is Cauchy for every  $j = 1, \dots, m$ .

*Go to proof.*

**Theorem 0.9.9** (Euclidean Cauchy Sequences Converge). *Every Cauchy sequence in  $\mathbb{R}^m$  is convergent in  $\mathbb{R}^m$ .*

*Go to proof.*

**Theorem 0.9.10** (Bolzano–Weierstrass in  $\mathbb{R}^m$ ). *Every bounded sequence in  $\mathbb{R}^m$  contains a subsequence that converges in  $\mathbb{R}^m$ .*

*Go to proof.*

## Proofs

# Topology

## 0.10 Topology

## 0.11 Definitions

**Definition 0.11.1** (Topological Space). Let  $X$  be a set. A *topology* on  $X$  is a collection  $\tau \subseteq \mathcal{P}(X)$  such that:

1.  $\emptyset \in \tau$  and  $X \in \tau$ ;
2. if  $\{U_i\}_{i \in I}$  is any family of sets in  $\tau$ , then  $\bigcup_{i \in I} U_i \in \tau$ ;
3. if  $U_1, \dots, U_n \in \tau$ , then  $\bigcap_{k=1}^n U_k \in \tau$ .

A *topological space* is a pair  $(X, \tau)$ , where  $X$  is a set and  $\tau$  is a topology on  $X$ . The members of  $\tau$  are called the *open sets* of the space.

**Definition 0.11.2** (Discrete Topology). Let  $X$  be a set. The *discrete topology* on  $X$  is the topology

$$\tau_{\text{disc}} = \mathcal{P}(X).$$

Thus every subset of  $X$  is open in the topological space  $(X, \tau_{\text{disc}})$ .

**Definition 0.11.3** (Sierpinski Topology). Let  $X = \{0, 1\}$ . The *Sierpinski topology* on  $X$  is the topology

$$\tau_S = \{\emptyset, \{1\}, X\}.$$

The topological space  $(X, \tau_S)$  is called the *Sierpinski space*.

**Definition 0.11.4** (Cofinite Topology). Let  $X$  be a set. The *cofinite topology* on  $X$  is the topology

$$\tau_{\text{cof}} = \{U \subseteq X : X \setminus U \text{ is finite}\} \cup \{\emptyset\}.$$

Thus a subset of  $X$  is open exactly when it is empty or its complement in the ambient set is finite.

**Definition 0.11.5** (Trivial Topology). Let  $X$  be a set. The *trivial topology* on  $X$  is the topology

$$\tau_{\text{triv}} = \{\emptyset, X\}.$$

Thus only  $\emptyset$  and  $X$  are open in the topological space  $(X, \tau_{\text{triv}})$ .

## 0.12 Topological Space Subsets

**Definition 0.12.1** (Closed Subset of a Topological Space). Let  $(X, \tau)$  be a topological space, and let  $F \subseteq X$ . The subset  $F$  is *closed in*  $(X, \tau)$  if its complement

$$X \setminus F$$

is open in  $(X, \tau)$ .

**Theorem 0.12.2** (Closed-Set Characterization of Topologies). *Let  $X$  be a set, and let  $\mathcal{C}$  be a collection of subsets of  $X$ . Suppose that:*

1.  $\emptyset \in \mathcal{C}$  and  $X \in \mathcal{C}$ ;
2. the intersection of any family of elements of  $\mathcal{C}$  belongs to  $\mathcal{C}$ ;
3. the union of any two elements of  $\mathcal{C}$  belongs to  $\mathcal{C}$ .

*Then there is a unique topology  $\tau$  on  $X$  whose family of closed sets is exactly  $\mathcal{C}$ .*

*Go to proof.*

**Definition 0.12.3** (Finer and Coarser Topologies). Let  $\tau_1$  and  $\tau_2$  be topologies on a set  $X$ . The topology  $\tau_1$  is *finer than*  $\tau_2$  if

$$\tau_2 \subseteq \tau_1.$$

Equivalently,  $\tau_2$  is *coarser than*  $\tau_1$ . The two topologies are *comparable* if one is finer than the other.

## 0.13 Basis

**Definition 0.13.1** (Basis for a Topology). Let  $(X, \tau)$  be a topological space. A collection  $\mathcal{B} \subseteq \tau$  is a *basis for the topology  $\tau$*  if every open set  $O \in \tau$  is the union of some subcollection of  $\mathcal{B}$ . The elements of  $\mathcal{B}$  are called *basic elements* or *basic open sets*.

**Proposition 0.13.2** (Pointwise Characterization of a Basis). *Let  $(X, \tau)$  be a topological space, and let  $\mathcal{B} \subseteq \tau$ . Then  $\mathcal{B}$  is a basis for  $\tau$  if and only if for every open set  $O \in \tau$  and every  $x \in O$ , there is  $B \in \mathcal{B}$  such that*

$$x \in B \subseteq O.$$

*Consequently, a subset  $A \subseteq X$  is open if and only if for each  $x \in A$ , there is  $B \in \mathcal{B}$  such that*

$$x \in B \subseteq A.$$

*Go to proof.*

## Proofs

# Set Algebra

## 0.14 Set Algebras

## 0.15 Systems of Sets

### 0.15.1 Systems of Sets

**Definition 0.15.1** (Semiring of Sets). A collection  $\mathcal{S}$  of subsets of a set  $X$  is a *semiring* if:

- (i)  $\emptyset \in \mathcal{S}$ ;
- (ii)  $A, B \in \mathcal{S}$  implies  $A \cap B \in \mathcal{S}$ ;
- (iii) for any  $A, B \in \mathcal{S}$ , the difference  $A \setminus B$  is a *finite disjoint union* of members of  $\mathcal{S}$ : there exist pairwise disjoint  $C_1, \dots, C_n \in \mathcal{S}$  with  $A \setminus B = C_1 \sqcup \dots \sqcup C_n$ .

**Definition 0.15.2** (Ring of Sets). A collection  $\mathcal{R}$  of subsets of  $X$  is a *ring of sets* if:

- (i)  $\emptyset \in \mathcal{R}$ ;
- (ii)  $A, B \in \mathcal{R}$  implies  $A \cup B \in \mathcal{R}$ ;
- (iii)  $A, B \in \mathcal{R}$  implies  $A \setminus B \in \mathcal{R}$ .

**Definition 0.15.3** (Algebra of Sets). A collection  $\mathcal{A}$  of subsets of  $X$  is an *algebra of sets* (or *Boolean algebra of sets*, or *field of sets*) if:

- (i)  $X \in \mathcal{A}$ ;
- (ii)  $A \in \mathcal{A}$  implies  $A^c = X \setminus A \in \mathcal{A}$ ;
- (iii)  $A, B \in \mathcal{A}$  implies  $A \cup B \in \mathcal{A}$ .

**Definition 0.15.4** ( $\sigma$ -Algebra). A collection  $\mathcal{M}$  of subsets of  $X$  is a  *$\sigma$ -algebra* (or  *$\sigma$ -field*) on  $X$  if:

- (i)  $X \in \mathcal{M}$ ;
- (ii)  $A \in \mathcal{M}$  implies  $A^c \in \mathcal{M}$ ;
- (iii) if  $\{A_n\}_{n \in \mathbb{N}}$  is a countable family in  $\mathcal{M}$ , then  $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$ .

The pair  $(X, \mathcal{M})$  is called a *measurable space*.

**Definition 0.15.5** (Topology). A *topology* on a set  $X$  is a collection  $\tau$  of subsets of  $X$  satisfying:

- (i)  $\emptyset \in \tau$  and  $X \in \tau$ ;
- (ii) if  $\{U_\alpha\}_{\alpha \in I}$  is any family of sets in  $\tau$  (indexed by any set  $I$ ), then  $\bigcup_{\alpha \in I} U_\alpha \in \tau$ ;



(iii) if  $U_1, \dots, U_n \in \tau$  (finitely many), then  $U_1 \cap \dots \cap U_n \in \tau$ .

The pair  $(X, \tau)$  is a *topological space*. Members of  $\tau$  are called *open sets*.

**Theorem 0.15.6** (Intersection of  $\sigma$ -algebras is a  $\sigma$ -algebra). *Let  $\{\mathcal{M}_\alpha\}_{\alpha \in I}$  be any family of  $\sigma$ -algebras on  $X$  (indexed by any set  $I$ ). Then*

$$\mathcal{M} := \bigcap_{\alpha \in I} \mathcal{M}_\alpha$$

*is a  $\sigma$ -algebra on  $X$ . Go to proof.*

**Definition 0.15.7** (Generated  $\sigma$ -Algebra). Let  $\mathcal{F}$  be any collection of subsets of  $X$ . The  *$\sigma$ -algebra generated by  $\mathcal{F}$* , written  $\sigma(\mathcal{F})$ , is the smallest  $\sigma$ -algebra on  $X$  that contains  $\mathcal{F}$ :

$$\sigma(\mathcal{F}) := \bigcap \{ \mathcal{M} \mid \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{M} \}.$$

The collection  $\mathcal{F}$  is called a *generating family* for  $\sigma(\mathcal{F})$ .

**Definition 0.15.8** (Borel  $\sigma$ -Algebra). Let  $(X, \tau)$  be a topological space. The *Borel  $\sigma$ -algebra* on  $X$ , written  $\mathcal{B}(X)$ , is the  $\sigma$ -algebra generated by the open sets:

$$\mathcal{B}(X) := \sigma(\tau).$$

Members of  $\mathcal{B}(X)$  are called *Borel sets*.

## 0.16 The Power Set and Characteristic Functions

### 0.16.1 The Power Set and Characteristic Functions

#### Proofs

# Algebras of Sets

0.17 Set Operations and Hierarchy

0.18 Boolean Algebra of Sets

0.19 Rings of Sets

0.20 Sigma Algebras

0.21 Borel Sets

Proofs

# Measure Spaces

## Proofs